

S8 — Grey Matter

8.1 The Nervous System

Organisation of the Nervous System

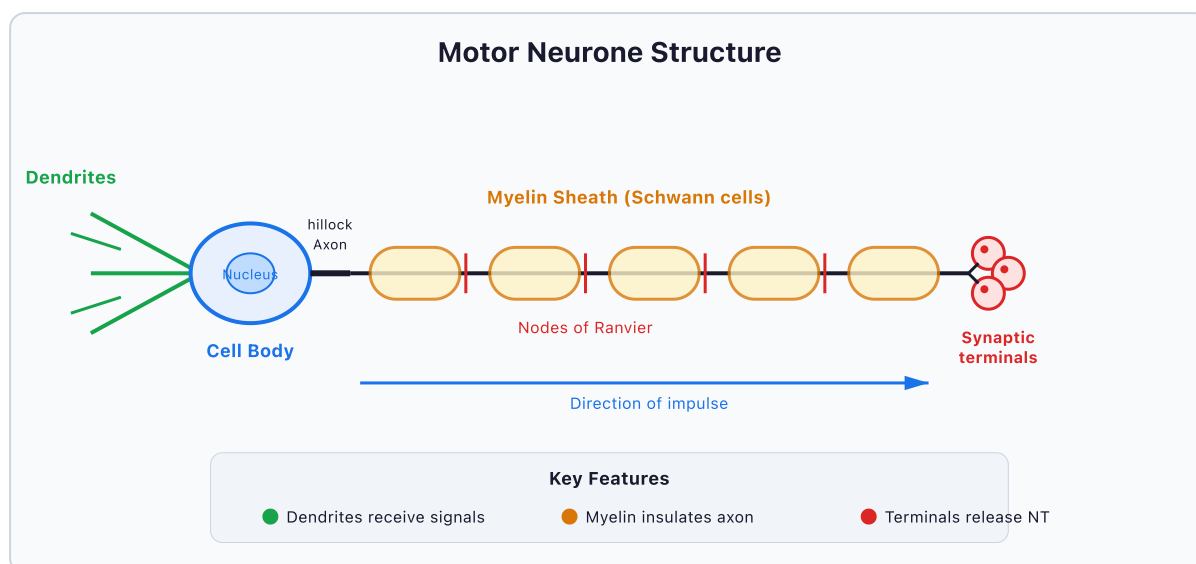
Nervous system: The organ system responsible for detecting stimuli, processing information, and coordinating rapid responses. It consists of the central nervous system (CNS: brain and spinal cord) and the peripheral nervous system (PNS: sensory and motor neurones).

The PNS is divided into the somatic nervous system (voluntary control of skeletal muscles) and the autonomic nervous system (involuntary control of smooth muscle, cardiac muscle, and glands). The autonomic system is further divided into the sympathetic (fight-or-flight) and parasympathetic (rest-and-digest) divisions, which generally have opposing effects.

Neurone Structure

Neurones (nerve cells) are specialised for rapid electrical signalling. They have:

- **Cell body (soma):** Contains the nucleus, many mitochondria, and rough ER (Nissl bodies) for protein synthesis.
- **Dendrites:** Short, branching extensions that receive impulses from other neurones or receptors.
- **Axon:** A long, thin extension that carries impulses away from the cell body. May be up to 1 metre long in motor neurones.
- **Myelin sheath:** Formed by Schwann cells wrapping around the axon (in the PNS). Insulates the axon, increasing the speed of impulse transmission. Gaps between Schwann cells are called nodes of Ranvier.
- **Axon terminal (synaptic knob):** Swollen end of the axon containing vesicles of neurotransmitter, mitochondria, and voltage-gated calcium channels.



Neurone Type	Function	Structure
Sensory	Carries impulses from receptors to CNS	Long dendron, short axon, cell body in dorsal root ganglion
Motor	Carries impulses from CNS to effectors	Short dendrites, long axon, cell body in CNS
Relay (interneurone)	Connects sensory and motor neurones in CNS	Short fibres, many connections

Responding to a stimulus: a receptor detects a stimulus and generates impulses in a sensory neurone, which carries them to the CNS; there, relay neurones pass the impulses to motor neurones, which carry them to an effector (a muscle that contracts or a gland that secretes). In a **reflex arc** (e.g. withdrawing the hand from a hot object), the pathway passes through the spinal cord via a relay neurone without involving conscious parts of the brain, so the response is rapid and automatic.

The Action Potential

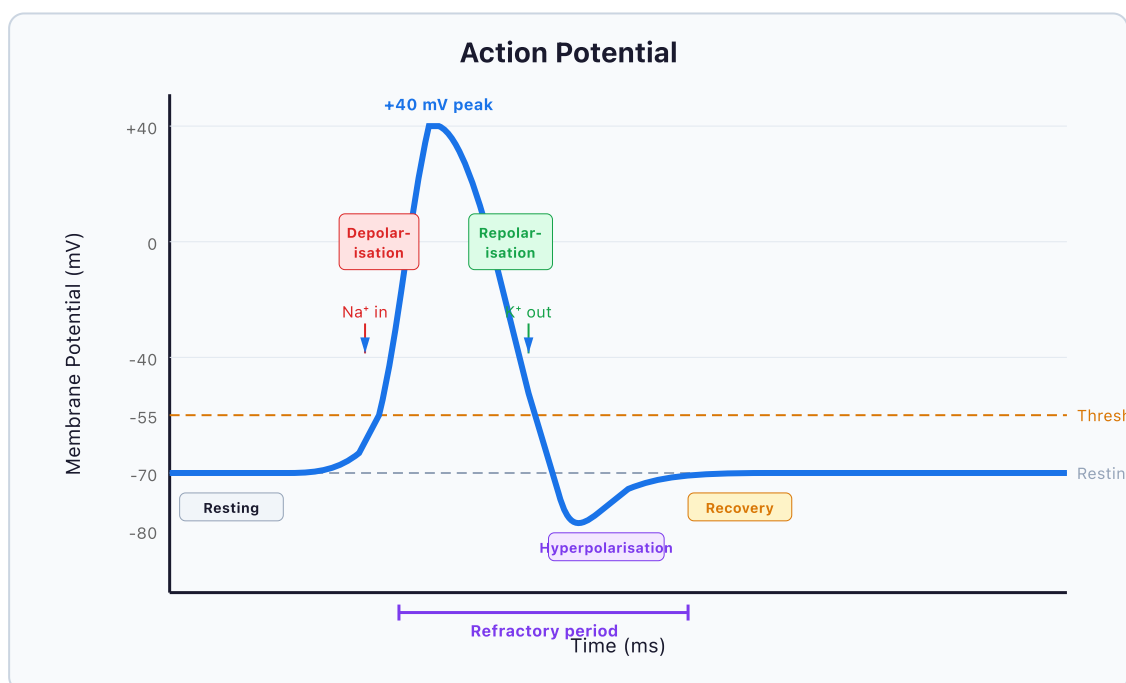
At rest, a neurone has a resting potential of approximately -70 mV (inside negative relative to outside), maintained by the sodium-potassium pump (3 Na^+ out, 2 K^+ in) and potassium leak channels.

- 1. Depolarisation:** A stimulus opens sodium voltage-gated channels. Na^+ ions rush in (down their electrochemical gradient), depolarising the membrane. If the threshold (-55 mV) is reached, more Na^+ channels open (positive feedback), causing rapid depolarisation to approximately $+40$ mV.

2. **Repolarisation:** Na^+ channels close (inactivate) and K^+ voltage-gated channels open. K^+ ions flow out, returning the membrane potential toward resting level.
3. **Hyperpolarisation:** K^+ channels close slowly, so the membrane briefly overshoots to about -80 mV .
4. **Recovery:** The Na^+/K^+ pump restores ion concentrations. During the refractory period, the neurone cannot fire again, ensuring unidirectional impulse propagation and limiting firing frequency.

Action potentials are **all-or-nothing**: if threshold is reached, a full action potential fires at maximum amplitude; if not, nothing happens. There is no such thing as a "small" or "large" action potential. Stimulus intensity is coded by the **frequency** of action potentials (a stronger stimulus produces more action potentials per second), not their size. The type of stimulus is coded by which neurones are firing (e.g., stimulation of the optic nerve is always interpreted as light, regardless of what caused it).

The refractory period has two phases: the **absolute refractory period** (Na^+ channels are inactivated and cannot reopen; no stimulus, however strong, can trigger another action potential) and the **relative refractory period** (some Na^+ channels have reset; a stronger-than-normal stimulus can trigger an action potential). The refractory period ensures: (1) unidirectional propagation (the region behind the action potential cannot be re-stimulated), (2) discrete action potentials (they do not merge), and (3) a maximum firing frequency (about 500-1000 Hz).



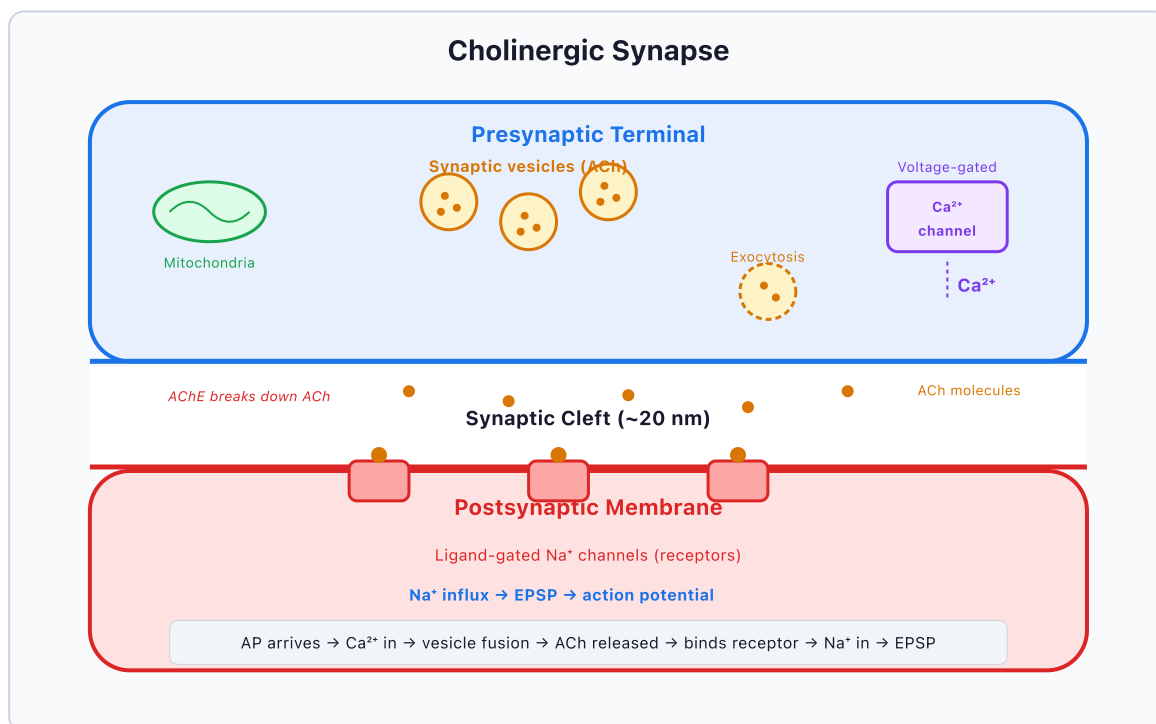
Saltatory Conduction

In myelinated neurones, the myelin sheath insulates the axon, preventing ion flow except at nodes of Ranvier. The action potential jumps from node to node (saltatory conduction), greatly increasing the speed of transmission (up to 120 m/s in large myelinated fibres vs 0.5 m/s in unmyelinated fibres). Speed is also increased by larger axon diameter (less internal resistance) and higher temperature (up to a point).

8.2 Synaptic Transmission

Structure of a Synapse

Synapse: A junction between two neurones (or a neurone and an effector cell) across which nerve impulses are transmitted chemically via neurotransmitters. The synaptic cleft is the narrow gap (~20 nm) between the presynaptic and postsynaptic membranes.



Mechanism of Synaptic Transmission

1. An action potential arrives at the presynaptic terminal (synaptic knob).
2. Depolarisation opens voltage-gated Ca^{2+} channels; Ca^{2+} ions flow into the synaptic knob.

3. Ca^{2+} causes synaptic vesicles (containing neurotransmitter, e.g., acetylcholine) to fuse with the presynaptic membrane and release their contents into the synaptic cleft by exocytosis.
4. Neurotransmitter diffuses across the cleft and binds to specific receptors on the postsynaptic membrane.
5. Receptor binding opens ligand-gated ion channels (e.g., Na^+ channels), causing depolarisation (excitatory postsynaptic potential, EPSP) or hyperpolarisation (inhibitory postsynaptic potential, IPSP) of the postsynaptic cell.
6. If enough EPSPs summate to reach threshold, an action potential is generated in the postsynaptic neurone.
7. Neurotransmitter is rapidly removed from the cleft by enzymatic breakdown (e.g., acetylcholinesterase breaks down acetylcholine), reuptake into the presynaptic terminal, or diffusion away. This prevents continuous stimulation.

Summation

Spatial summation: Multiple presynaptic neurones release neurotransmitter simultaneously onto the same postsynaptic neurone, and their combined EPSPs reach threshold.

Temporal summation: A single presynaptic neurone fires repeatedly in rapid succession, and the EPSPs accumulate to reach threshold before they decay.

Exam Tip: Synapses ensure unidirectional transmission (vesicles only in presynaptic terminal, receptors only on postsynaptic membrane). They also allow integration of information (summation), amplification (one vesicle contains thousands of neurotransmitter molecules), and can be modified (learning, drugs).

Coordination: Nervous, Hormonal and Plant Responses

Nervous and Hormonal Coordination in Animals

Feature	Nervous control	Hormonal control
Signal	Electrical impulses along neurones; neurotransmitters at synapses	Chemical hormones carried in the blood

Speed	Very rapid	Slower
Duration	Short-lived	Often long-lasting
Target	Specific, localised effectors	Any cells with specific receptors; often widespread

The two systems work together. For example, in the fight-or-flight response sympathetic neurones act on the heart immediately, while the adrenal medulla releases adrenaline into the blood to sustain the response throughout the body. The hypothalamus links the two systems by controlling the pituitary gland.

Plant Responses: Phytochrome and IAA

Plants have no nervous system and respond to environmental cues using chemicals that alter gene transcription.

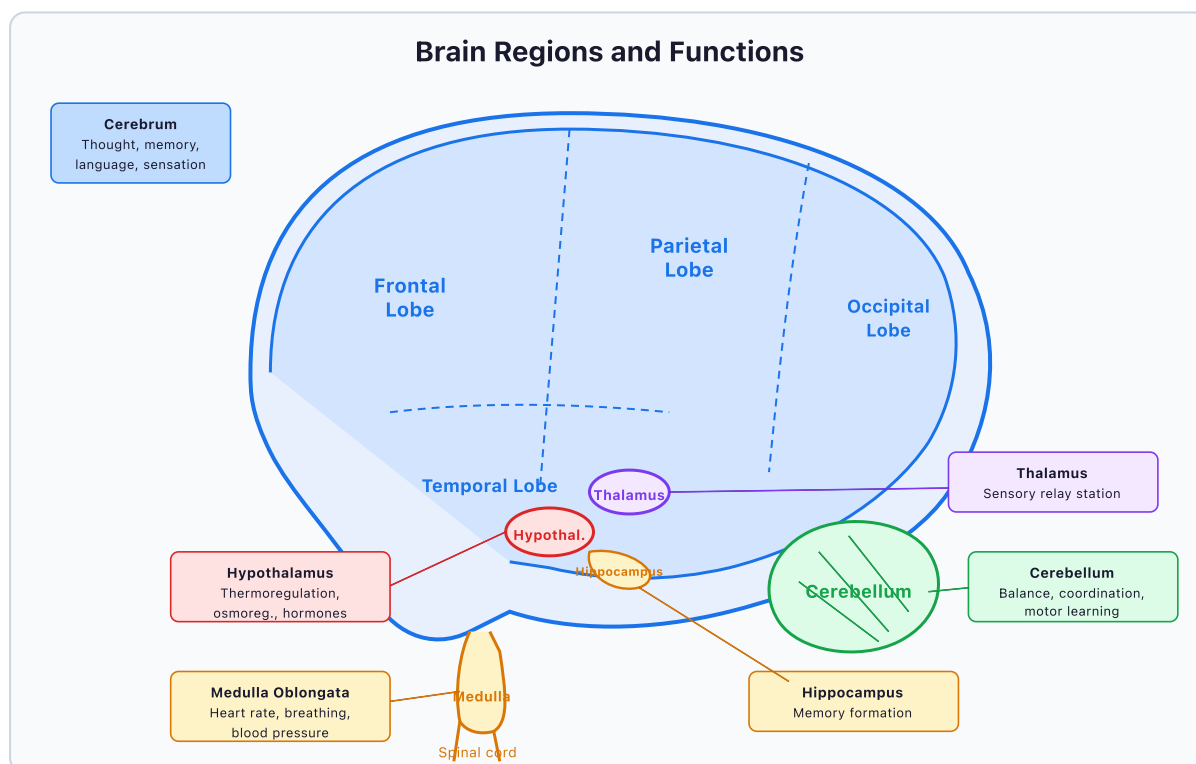
Phytochrome is a light-sensitive pigment that exists in two interconvertible forms. P_R (P_{660}) absorbs red light and is converted to P_{FR} (P_{730}); P_{FR} absorbs far-red light and is converted back to P_R , and P_{FR} also slowly reverts to P_R in the dark. P_{FR} is the physiologically active form: it controls responses such as seed germination, flowering (in response to day length) and greening of seedlings. P_{FR} moves into the nucleus and interacts with transcription factors, switching on or off the transcription of genes involved in these responses.

IAA (indoleacetic acid, an auxin) is produced in shoot tips and moves down the shoot, where it stimulates cell elongation. When light comes from one side, IAA accumulates on the shaded side, so cells there elongate more and the shoot bends towards the light (positive phototropism). IAA causes proton pumping into the cell wall; the low pH loosens the cell wall so the cell can expand. IAA also affects transcription: it triggers breakdown of repressor proteins, freeing transcription factors to switch on genes involved in growth.

8.3 The Brain

Brain Structure and Function

Region	Function
Cerebrum (cerebral cortex)	Higher functions: conscious thought, reasoning, language, voluntary movement, sensory perception, memory, personality. Divided into two hemispheres with specialised lobes (frontal, parietal, temporal, occipital).
Cerebellum	Coordination of voluntary movement, balance, posture, motor learning. Receives input from proprioceptors, eyes, and ears.
Medulla oblongata	Autonomic functions: heart rate, breathing rate, blood pressure, swallowing, vomiting. Contains cardiovascular and respiratory centres.
Hypothalamus	Homeostasis: thermoregulation, osmoregulation, hunger/thirst, circadian rhythms. Links nervous and endocrine systems via the pituitary gland.
Thalamus	Relay station: processes and routes sensory information to appropriate cortical areas.
Hippocampus	Formation of new long-term memories; spatial navigation.



Methods of Studying the Brain

- **fMRI (functional magnetic resonance imaging):** Detects changes in blood oxygenation to map active brain regions during tasks. Non-invasive, good spatial resolution, but temporal resolution is limited and it shows correlation not causation.
- **CT/MRI scans:** Structural imaging showing anatomy, tumours, damage. MRI provides better soft tissue contrast than CT.
- **How they work:** CT uses many thin X-ray beams from different angles to build up cross-sectional images; it shows dense structures and bleeding clearly (e.g. after a stroke). MRI uses a strong magnetic field and radio waves and gives detailed images of soft tissue, e.g. to locate tumours. fMRI works on the same principle but detects the higher proportion of oxyhaemoglobin in blood flowing to active brain regions.
- **PET (positron emission tomography):** A radioactive tracer (e.g. radiolabelled glucose) is injected; active brain areas take up more tracer, and the positrons it emits are detected to build images of brain activity. Used to investigate brain function and diagnose conditions such as Alzheimer's disease, tumours and epilepsy.
- **EEG (electroencephalography) — extension, not named in 9BN0:** Records electrical activity across the scalp. Good temporal resolution; used to study sleep, epilepsy. Poor spatial resolution.
- **Studies of brain damage:** Patients with specific lesions reveal function of damaged areas (e.g., Broca's area damage causes speech production problems).
- **Electrical stimulation:** Direct stimulation during surgery can map motor and sensory cortex.

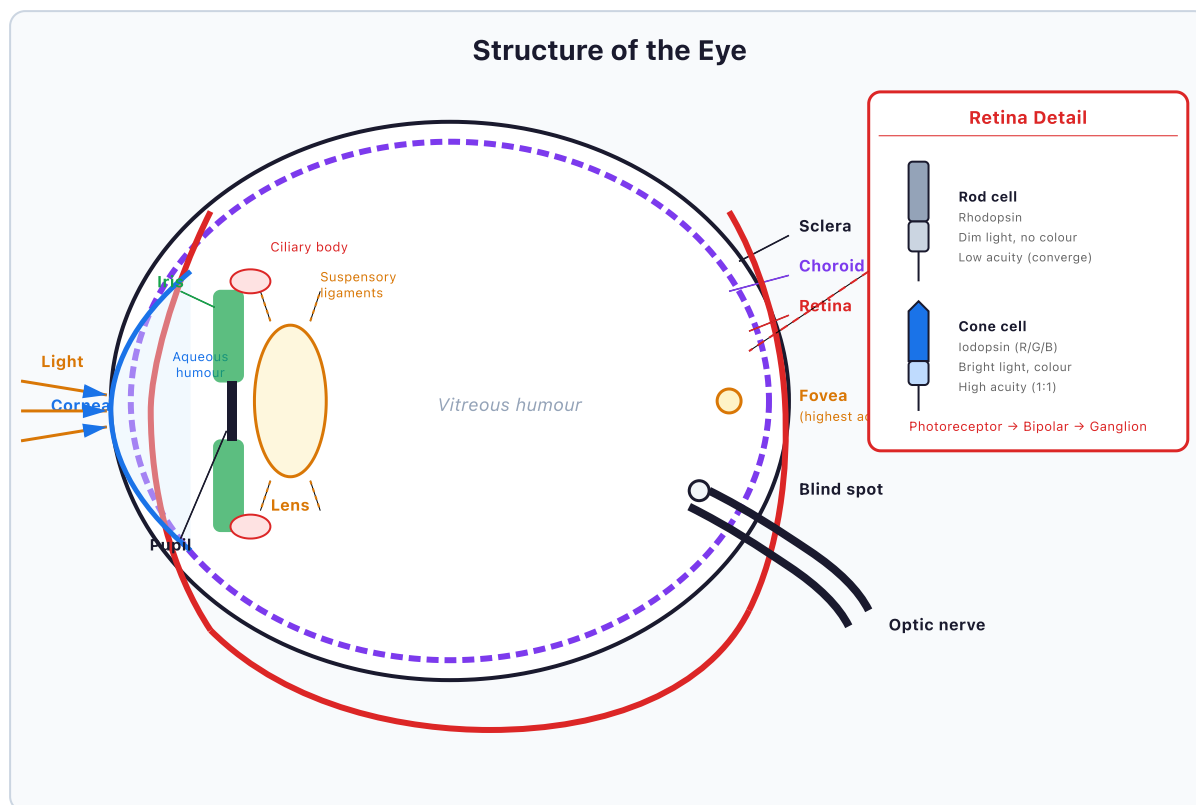
8.4 Vision

Structure of the Eye

The human eye focuses light onto the retina, which contains photoreceptors that convert light into electrical signals. Key structures include the cornea (most refraction), iris (controls pupil size and light entry), lens (fine focusing by accommodation), vitreous humour (maintains eye shape), and retina (contains photoreceptors).

The pupil reflex: the iris contains two antagonistic sets of muscles: **circular muscles** (supplied by parasympathetic neurones) and **radial muscles** (supplied by sympathetic neurones). In **bright light**, photoreceptors in the retina send impulses along the optic nerve to the brain, which sends impulses along parasympathetic motor neurones to the iris: the circular muscles contract and the radial muscles relax, so the pupil **constricts**, protecting the retina. In **dim light**, sympathetic neurones stimulate the

radial muscles to contract while the circular muscles relax, so the pupil **dilates**, letting more light reach the retina.



Photoreceptors

Feature	Rod Cells	Cone Cells
Sensitivity	Very sensitive (work in dim light)	Less sensitive (need bright light)
Visual acuity	Low (many rods converge onto one bipolar cell — retinal convergence)	High (one cone per bipolar cell in fovea)
Colour vision	No (only contain rhodopsin; monochromatic)	Yes (three types with different opsins for red, green, blue)
Distribution	Mostly peripheral retina	Concentrated in fovea
Pigment	Rhodopsin (bleached by light; regenerated in dark)	Iodopsin (three types)

In the dark: cation channels in the rod cell membrane are open, so Na^+ ions diffuse in (and are pumped out of the inner segment), keeping the rod cell slightly depolarised (about -40 mV). The depolarised rod releases a neurotransmitter (glutamate) that inhibits the bipolar cell, so no action potentials are produced in the optic neurones.

In the light: rhodopsin absorbs light and breaks down (bleaches) into **retinal** and **opsin**. The opsin activates a cascade of reactions that closes the cation channels. Na^+ ions are still pumped out but no longer enter, so the rod cell becomes **hyperpolarised** (about -70 mV) and stops releasing neurotransmitter. The bipolar cell is no longer inhibited, so it depolarises and releases neurotransmitter onto a ganglion cell, generating action potentials in the optic neurone (whose axons form the optic nerve). Rhodopsin is resynthesised from retinal and opsin in the dark, using ATP.

The Critical Period and Visual Development

There is a critical period in early development during which visual experience is essential for normal development of the visual cortex. Lack of stimulation during this period (e.g., due to cataracts or strabismus) can lead to permanent visual impairment (amblyopia) even if the physical defect is later corrected. This demonstrates neural plasticity during development — connections that are used are strengthened while unused connections are pruned.

Hubel and Wiesel's Experiments

Hubel and Wiesel recorded from neurones in the visual cortex of kittens and monkeys. In normal animals, axons from the two eyes are sorted into alternating **ocular dominance columns** in the visual cortex, one set for each eye. When they stitched one eye closed in newborn animals during the critical period, then reopened it months later, the animal was blind in that eye: the columns for the open eye had spread and taken over most of the cortex, while those for the deprived eye had shrunk. Closing an eye in adult animals had no effect. This showed that visual stimulation during the critical period is needed for the visual cortex to develop, and that axons from the two eyes compete for synapses (active synapses are kept; inactive ones are lost).

Animal Models in Research and Ethics

Animal models have been used in brain research because their nervous systems are similar to ours (e.g. cats and monkeys for vision) and experiments that would be unethical in humans can be carried out. Two ethical standpoints on using animals:

- **Rights-based (absolutist) standpoint:** animals have rights, so causing them suffering is wrong whatever the benefits; animal experiments are never acceptable.
- **Utilitarian standpoint:** an action is right if it produces the greatest good for the greatest number; animal experiments are acceptable if the benefits (e.g. treatments for human disease) outweigh the suffering caused.

In the UK, animal research is regulated by law and follows the **3Rs: replacement** (use alternatives where possible), **reduction** (use as few animals as possible) and **refinement** (minimise suffering).

8.5 Learning, Memory, and Habituation

Types of Learning

Learning: A relatively permanent change in behaviour as a result of experience. It involves changes in the structure and function of synapses (synaptic plasticity).

- **Habituation:** A reduction in response to a repeated, harmless stimulus. The organism learns to ignore irrelevant stimuli, conserving energy. Involves a decrease in neurotransmitter release at synapses in the reflex pathway (fewer Ca^{2+} channels open, less vesicle fusion).
- **Classical conditioning (Pavlovian) — extension, not in 9BN0:** Associating a neutral stimulus with an unconditioned stimulus until the neutral stimulus alone produces the response (conditioned response). Example: Pavlov's dogs salivating at the sound of a bell.
- **Operant conditioning — extension, not in 9BN0:** Learning through reinforcement (reward) or punishment. Behaviour followed by positive consequences is repeated; behaviour followed by negative consequences is reduced.

Core Practical 18: Investigate Habituation to a Stimulus

1. Place a snail on a clean, damp surface and allow it to acclimatise until its tentacles are fully extended.
2. Gently touch the snail between the eye stalks with a damp cotton bud (the stimulus) and start a stopwatch. Record the time taken for the tentacles to fully re-extend.
3. Repeat the stimulus as soon as the tentacles re-emerge (or at fixed intervals), for about 10–20 touches.

4. Control variables: same snail, same strength and position of touch, same interval between stimuli, same temperature and light.

With repeated touches, the time taken for the tentacles to re-emerge decreases: the snail habituates.

Ethics: handle the snail gently, keep it damp, and return it to its habitat afterwards.

Memory (extension: not in the 9BN0 specification)

Short-term memory (STM): Limited capacity (~7 items), short duration (seconds to minutes).

Maintained by reverberating circuits — repeated firing of neurones in closed loops.

Long-term memory (LTM): Potentially unlimited capacity and duration. Requires consolidation — the transfer of information from STM to LTM. This involves structural changes at synapses (long-term potentiation, LTP): formation of new synaptic connections, increased receptor density, increased neurotransmitter release, and growth of new dendritic spines. The hippocampus is critical for forming new declarative (factual) memories.

Long-Term Potentiation (LTP) (extension: not in the 9BN0 specification)

LTP is a long-lasting increase in synaptic strength following high-frequency stimulation. It is considered the cellular basis of learning and memory. Repeated stimulation leads to increased neurotransmitter release, more postsynaptic receptors (especially NMDA and AMPA glutamate receptors), and structural changes that make the synapse more efficient. This is summarised as "neurones that fire together, wire together" (Hebb's rule).

8.6 Drugs and the Brain

How Drugs Affect Synapses

Psychoactive drugs alter brain function by modifying synaptic transmission. They can act as:

- **Agonists:** Mimic the neurotransmitter and bind to receptors, producing the same effect (e.g., nicotine mimics acetylcholine).
- **Antagonists:** Block receptors, preventing the neurotransmitter from binding (e.g., curare blocks acetylcholine receptors at neuromuscular junctions).
- **Reuptake inhibitors:** Block reuptake of neurotransmitter, prolonging its effect in the synapse (e.g., SSRIs block serotonin reuptake; cocaine blocks dopamine reuptake).

- **Enzyme inhibitors:** Block enzymes that break down neurotransmitters, increasing their concentration (e.g., nerve gases inhibit acetylcholinesterase).
- **Release stimulators:** Increase neurotransmitter release from presynaptic terminals (e.g., amphetamines increase dopamine release).

Dopamine and Addiction (extension: not in the 9BN0 specification)

The brain's reward pathway (mesolimbic pathway) uses dopamine as a neurotransmitter. Activities that promote survival (eating, social bonding) naturally stimulate dopamine release, creating feelings of pleasure and reinforcing the behaviour. Addictive drugs hijack this system by flooding synapses with dopamine, producing intense pleasure and reinforcing drug-taking behaviour. Over time, the brain downregulates dopamine receptors (tolerance), requiring more of the drug for the same effect, and creating withdrawal symptoms when the drug is absent.

Serotonin and Depression

Depression is associated with low levels of the neurotransmitter **serotonin** in the brain (e.g. reduced serotonin production or fewer serotonin receptors), so fewer impulses are transmitted in the pathways involved in mood. Understanding these imbalances in naturally occurring brain chemicals has led to the development of new drugs: for example, selective serotonin reuptake inhibitors (SSRIs, e.g. fluoxetine) block the reuptake of serotonin into the presynaptic neurone, so serotonin stays in the synaptic cleft longer and stimulates the postsynaptic neurone more. Similarly, knowing that Parkinson's disease involves a lack of dopamine led to L-Dopa and dopamine agonists.

MDMA (Ecstasy)

MDMA binds to the serotonin reuptake transporter proteins in the presynaptic membrane, inhibiting serotonin reuptake and causing the transporters to work in reverse, releasing more serotonin into the synaptic cleft. The raised serotonin concentration in the cleft keeps postsynaptic neurones stimulated, causing the mood-elevating effects. MDMA can also increase dopamine release. Repeated use can deplete serotonin, which may contribute to low mood afterwards.

8.7 Neurodegenerative Diseases

Parkinson's Disease

Parkinson's disease: A progressive neurodegenerative disorder caused by the death of dopamine-producing neurones in the substantia nigra (a region of the midbrain). Loss of dopamine disrupts the basal ganglia circuits that control movement.

Symptoms include tremor at rest, rigidity (stiff muscles), bradykinesia (slow movement), postural instability, and eventually cognitive decline. Treatment includes L-DOPA (a dopamine precursor that can cross the blood-brain barrier and be converted to dopamine in the brain), dopamine agonists, and deep brain stimulation. L-DOPA becomes less effective over time and can cause side effects.

How L-Dopa affects synaptic transmission: dopamine itself cannot cross the blood-brain barrier, but L-Dopa can. In the brain it is converted to dopamine by the enzyme dopa decarboxylase, increasing dopamine synthesis and release by the surviving neurones, so more dopamine binds to postsynaptic receptors and control of movement improves.

Alzheimer's Disease (extension: not in the 9BN0 specification)

Alzheimer's disease: A progressive neurodegenerative disease and the most common cause of dementia. Characterised by the formation of amyloid plaques (aggregates of beta-amyloid protein between neurones) and neurofibrillary tangles (twisted tau protein filaments inside neurones) that lead to neuronal death.

The disease primarily affects the hippocampus and cerebral cortex. Early symptoms include memory loss (especially recent events), confusion, and difficulty with language. As the disease progresses, patients lose the ability to perform daily tasks, experience personality changes, and eventually lose recognition of family members. There is no cure; current treatments (acetylcholinesterase inhibitors such as donepezil) only temporarily slow symptom progression by increasing acetylcholine levels in the brain.

Risk factors include age (strongest risk factor), genetics (APOE4 allele, familial mutations in APP, presenilin genes), head injuries, cardiovascular disease risk factors, and possibly environmental factors. Research into disease-modifying treatments (anti-amyloid antibodies, tau-targeting therapies) is ongoing.

Exam Tip: When comparing Parkinson's and Alzheimer's, note the key differences: Parkinson's primarily affects movement (dopamine loss in substantia nigra), while Alzheimer's primarily affects

memory and cognition (widespread cortical neurone loss with plaques and tangles). Both are progressive and currently incurable, but their mechanisms, symptoms, and treatments differ significantly.

8.8 Genomes, Genetic Modification and Brain Development

Genome Sequencing and Personalised Medicine

Genome sequencing projects (such as the Human Genome Project) have identified human genes and many variations in them, e.g. single nucleotide polymorphisms (SNPs). Some variations affect how a person responds to a drug, for example how quickly a drug is metabolised. **Personalised medicine** uses a person's genotype to choose the most effective drug and dose with the fewest side effects, and to predict the risk of particular diseases. Social, moral and ethical issues include: who has access to the information (insurers, employers) and the risk of discrimination, privacy and consent, anxiety caused by knowing one's disease risk, and the cost and fair availability of personalised treatments.

Producing Drugs Using Genetically Modified Organisms

- **Microorganisms:** the human gene (e.g. for insulin) is cut out with restriction enzymes and inserted into a plasmid vector using DNA ligase. The recombinant plasmid is taken up by bacteria, which are grown in fermenters, and the protein is extracted and purified.
- **Plants:** the gene is inserted into plant cells, e.g. using the Ti plasmid of *Agrobacterium tumefaciens* as a vector, and whole plants are grown from the modified cells; the plants make the drug (e.g. vaccines or antibodies).
- **Animals ("pharming"):** the gene is joined to a promoter for a milk protein and microinjected into a fertilised egg, which is implanted into a surrogate mother; the transgenic animal secretes the drug in its milk (e.g. human antithrombin from goats).

Risks and Benefits of Genetically Modified Organisms

Benefits	Risks
Large quantities of pure human proteins (e.g. insulin) produced relatively cheaply, with no rejection or disease risk from animal sources	Genes may spread to wild relatives (e.g. herbicide-resistant "superweeds")

Crops with higher yield, pest resistance (less insecticide use) or herbicide resistance	Harm to non-target organisms and reduced biodiversity
Crops with improved nutritional value (e.g. Golden Rice with extra β -carotene)	Antibiotic-resistance marker genes could transfer to bacteria; possible allergic reactions to new proteins
Crops that grow in poor conditions (e.g. drought tolerance)	Animal welfare concerns in pharming; dependence of farmers on seed companies

Nature and Nurture in Brain Development

- **New-born babies:** abilities present at birth (e.g. preferring to look at faces, reflexes) are likely to be innate (nature), since there has been little time to learn.
- **Animal experiments:** e.g. Hubel and Wiesel's deprivation experiments, or kittens reared seeing only vertical stripes that later fail to respond to horizontal lines, show that visual experience (nurture) is needed for normal development.
- **Individuals with damaged brain areas:** e.g. after a stroke or injury, the abilities lost show what that area normally does, and recovery shows the brain's plasticity.
- **Twin studies:** identical twins share all their genes and non-identical twins about half; comparing how similar each type is (especially twins reared apart) indicates the relative contribution of genes and environment.
- **Cross-cultural studies:** comparing people raised in different cultures (e.g. responses to visual illusions such as the Müller-Lyer illusion) shows the effect of environment; features shared by all cultures are more likely to be innate.